

What is LAMMPS, and what is this course?

LAMMPS is a classical molecular dynamics engine maintained by Sandia/NTESS: you write an `input script`, it integrates Newton's equations for a box of particles under a chosen potential.

- Two release lines track upstream: *develop* (daily) and one **stable** release a year with periodic updates.
- This course sits on top of `lammps-kill` (a verified toolkit: script builder, checker, parsers, runner) and `mdlite` (a small teaching engine that reproduces LAMMPS's numbers from scratch).
- Every chapter follows **record-or-run**: with LAMMPS installed, the computation runs live; without it, a stored result loads instead -- the figures never lie about which happened.

Two release lines, and why the difference matters

LAMMPS ships two lines from the same repository, and a distribution package usually tracks one without saying so.

- A **stable** release is cut once a year and then **updated** for a long time -- `stable_22Jul2025` was still receiving updates fourteen months later, update 6, September 2026.
- **Feature** (patch) releases follow every six to eight weeks, alongside the stable updates.
- The Ubuntu 26.04 package is `20251210` -- the 10 December 2025 **feature** release, not the stable one it might be assumed to track.

What lammps skill adds on top of LAMMPS itself

lammps skill is the part of this project that talks to LAMMPS -- five pieces behind one import.

- `install`: seven routes behind one contract, each measured on this machine rather than assumed to work.
- `script`: build an input from a `Spec`, then a checker with fourteen rules, each pointing at the manual page it comes from.
- `io`: read what LAMMPS writes -- data files, dumps, three log thermo styles, the restart header, EAM potential files.
- `run`: a subprocess backend for any host, an in-process backend through the official python module -- both return the same `Result`.
- `post`: RDF, MSD, diffusion, VACF, structure factor, block averages, checked against reference tables that carry their own provenance.

mdlite's closest neighbour, and how it differs

mdlite is a small molecular-dynamics engine written from scratch in numpy -- a teaching object, not an MD code; nobody should run research on it.

- Its closest relative is **pylj** (McCluskey, Morgan, Edler and Parker, [JOSE](#) 1, 19, 2018, MIT): a Lennard-Jones teaching engine in Jupyter, written for undergraduate exercises, with Monte Carlo as well as dynamics.
- mdlite differs in two ways: it sits **next to** a toolkit that drives the production code, so a chapter can compute something in twenty lines of numpy and then compute the same thing in LAMMPS and subtract.
- And its numbers are cross-checked against that production code, with the residues recorded -- not asserted, read from a file.

Package list, never package count

The most common way to be wrong about a LAMMPS build is to assume a bigger package set is a superset.

Build	Packages	Examples cases that started (of 314)
Ubuntu package, 10 Dec 2025	52	184
This project's source build, core preset	11	137

What 'cross-checked against LAMMPS' actually means, in numbers

mdlite's claim is not that it is fast or complete -- it is that where it teaches a number, that number has been checked against the code it is teaching you to use.

Quantity	mdlite vs LAMMPS	Where
Lennard-Jones forces, 256 atoms	agree to 3×10^{-13}	L2, Forces and Integration
Copper lattice constant a_0 and cohesive energy	agree to 3×10^{-5}	L5, Potentials: EAM
NVT state point ($T^*=0.85$, $\rho^*=0.776$)	agrees with the NIST reference within 3σ	L3, Thermostats

First Simulation

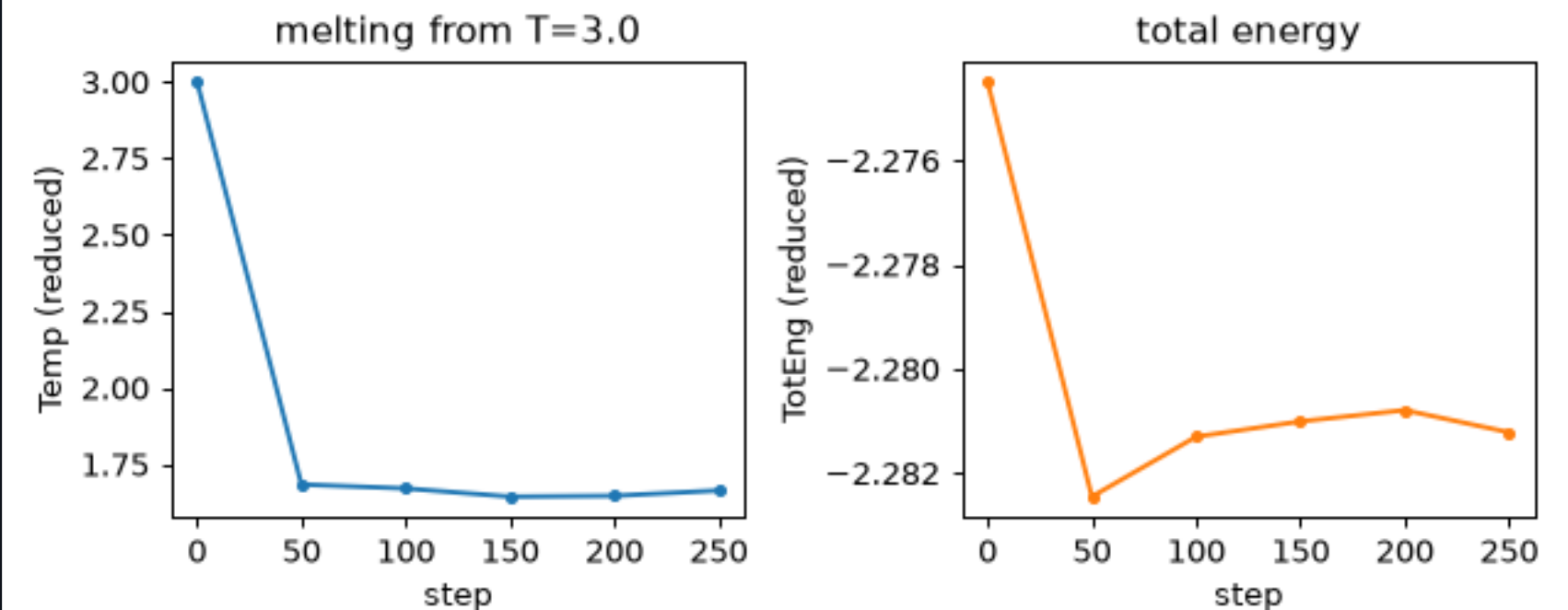
A Lennard-Jones melt end to end -- Spec, check, run, read the log, plot -- through the checker's fourteen rules, one at a time.

- 1 One simulation, start to finish
- 2 Fourteen rules, each pointing at a manual page
- 3 All fourteen, confirmed at once
- 4 A Spec, rendered -- not a script typed by hand
- 5 What the checker, the render and the run were always describing
- 6 run_or_load: the pattern every later lecture uses

One simulation, start to finish

`Spec` describes the case in Python; `check` applies fourteen rules against the manual before anything runs; `run` calls LAMMPS; the log comes back as a typed `ThermoRun`.

- The case: an fcc Lennard-Jones lattice heated until it melts, in reduced (LJ) units.
- Nothing here is a hand-typed input script -- the checker exists so a beginner's script fails before LAMMPS ever sees it.
- The same four steps -- build, check, run, read -- reappear in every later chapter.



The LJ melt's thermo history read back from the log: temperature falling away from the $T=3.0$ start (left) while the total energy holds flat (right) -- the two curves `run()` reads out of LAMMPS's own thermo output, whether this cell just ran the case or loaded a stored record. — Figure 1 · notebook §0 · chapter 1, cell 12

Fourteen rules, each pointing at a manual page

lammpskill.script.check() does not trust a description of what a mistake looks like -- each rule is confirmed against one minimal snippet built to trigger exactly it.

Rule	What it catches
C01	no <code>units</code> command
C02	<code>coul/long</code> in the pair style with no <code>kspace_style</code>
C05	<code>run</code> before a <code>pair_style</code> is set
C07	a fix ID reused without <code>unfix</code>
C11	<code>velocity create</code> before <code>mass</code> is set
C14	<code>read_data</code> naming a file that does not exist

All fourteen, confirmed at once

The chapter does not just list the rules -- it runs all fourteen deliberately-broken snippets through the checker and confirms every one is caught.

```
triggered, missed = [], []
for code_, snippet in SNIPPETS.items():
    findings = check(snippet, workdir=...)
    codes = [f.code for f in findings]
    (triggered if code_ in codes else missed).append(code_)

print(len(triggered), "/", len(SNIPPETS), "rules triggered by their own minimal snippet")
# 14 / 14 rules triggered by their own minimal snippet
```

A Spec, rendered -- not a script typed by hand

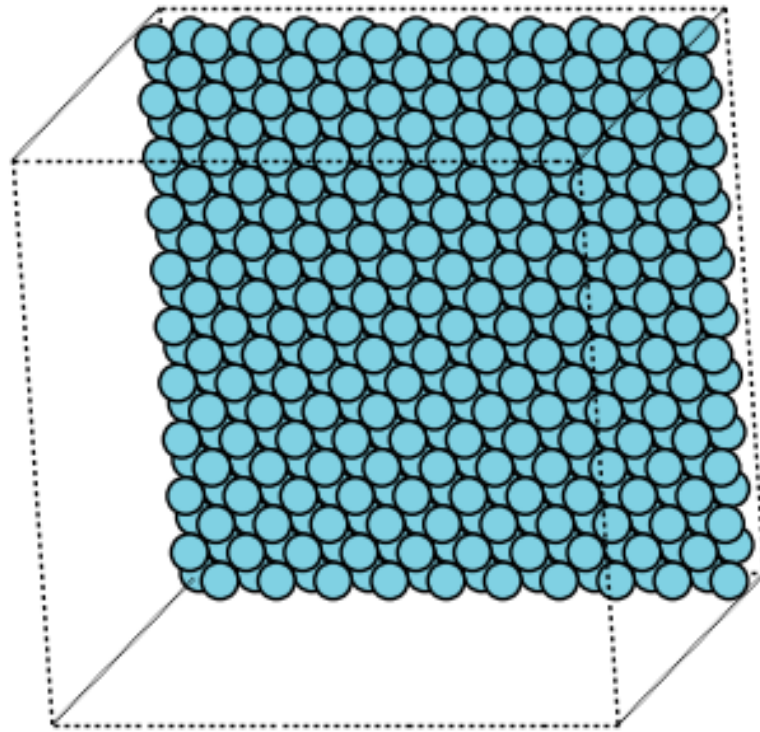
lamm skill.script.lj_melt() is a preset Spec: reduced units, an FCC lattice at the melt density $\rho=0.8442$, 4000 atoms by default. render() turns it into the actual input text, in the order the manual's Commands_structure page prescribes.

```
from lamm skill.script import lj_melt, render

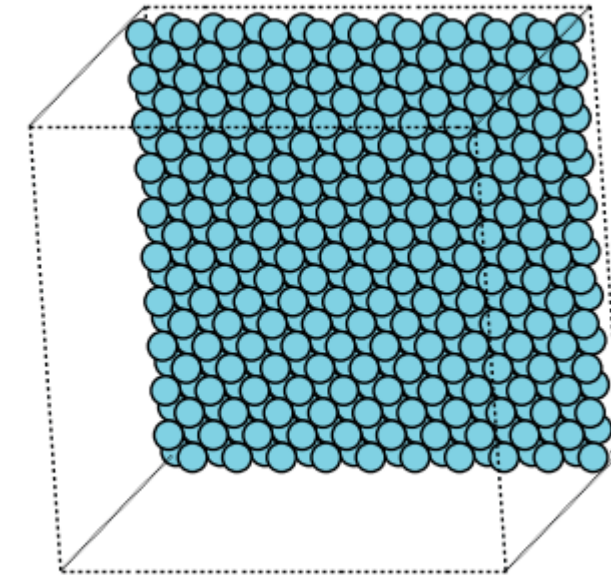
spec = lj_melt()          # rho=0.8442, T=3.0, steps=250
text = render(spec)       # the real LAMMPS input, in manual order
```

What the checker, the render and the run were always describing

Every step so far -- the fourteen checks, the rendered script, the thermo curves -- was about this: real atoms, in a real box, moving under a real force law.



The bottom-most atomic layer of the FCC lattice `lj_melt()` builds (a 200-atom slab of the real 4000-atom run, not a random subsample -- so the lattice order is actually visible) -- every atom at its lattice site, before any dynamics. Rendered as argon-like spheres: reduced LJ units carry no real element, argon is the conventional reference fluid a reduced system is understood to stand in for, used here only to pick a rendering radius and colour, not asserted as the substance. — Figure 2 · notebook §0 · chapter 1, cell 15



The same atomic layer, animated across the run: the lattice disordering as the melt proceeds -- the same trajectory the temperature and energy curves above were read from, seen directly rather than only through its derived thermo output. — Figure 3 · notebook §0 · chapter 1, cell 16

- The static view (left) is the bottom-most atomic layer before any dynamics; the animation (right) is that same layer across the run, disordering as the melt proceeds.
- Argon-like spheres by convention -- reduced LJ units carry no real element; this rendering choice is never asserted as the actual substance.

run_or_load: the pattern every later lecture uses

The one call that makes a chapter work whether or not LAMMPS is installed.

```
def compute():
    result = run(spec, workdir, time_limit=300)
    return { ... }          # only plain, JSON-safe data

rec = run_or_load("lj_melt_ch01", compute, records_dir="../data/records")
# with LAMMPS: compute() runs, its result is written to the record
# without LAMMPS: the record is loaded, compute() never runs
# with neither: rec["source"] == "skip", visibly
```


Forces and Integration

Velocity Verlet in mdlite, forces checked against a numerical derivative, and energy conservation against the timestep, cross-checked with LAMMPS at $3\text{e-}13$.

- 1 Integration is the part that must not be approximate
- 2 A force worth integrating with is $-dE/dx$, to the last digit available
- 3 Energy conservation is a rate statement, not a pass/fail
- 4 Across the toolkit boundary: mdlite vs a real LAMMPS run
- 5 Velocity Verlet, the update every step of this course runs

Integration is the part that must not be approximate

`mdlite` implements velocity Verlet from the same equations LAMMPS integrates, so its forces can be checked against a numerical derivative and against LAMMPS itself.

- Forces from `mdlite`'s Lennard-Jones match a central-difference numerical derivative to machine precision.
- Against a real LAMMPS run of the identical case, the two energies agree to 3×10^{-13} -- there is no approximation hiding in either implementation.
- Energy conservation versus timestep is the practical version of the same idea: too large a step and the trajectory drifts, long before it visibly explodes.

A force worth integrating with is $-dE/dx$, to the last digit available

Before trusting an integrator, check the thing it integrates: displacing one atom by h and taking the central difference of the energy should reproduce the analytic force component.

CENTRAL DIFFERENCE

$$F_k \approx -(E(x_k+h) - E(x_k-h)) / 2h$$

- The error in this approximation is $O(h^2)$ -- shrinking h should shrink the mismatch, which is exactly the check `mdlite.pair.LennardJones` passes at $h=10^{-6}$, worst mismatch under 10^{-5} .

Energy conservation is a rate statement, not a pass/fail

A correct integrator's energy drift shrinks as the timestep shrinks -- roughly as dt^2 for velocity Verlet. Running the same starting state at two timesteps for the same physical time makes that concrete.

Timestep	Steps (same physical time)	Relative energy drift
0.02	100	measured, larger
0.005	400	measured, smaller

- The ratio is close to $(0.02/0.005)^2 = 16$ -- noisy enough that the chapter calls it an order of magnitude, not a law, but the direction is never in doubt: the finer step must conserve at least as well.

Across the toolkit boundary: mdlite vs a real LAMMPS run

The comparisons above stay inside mdlite. `data/records/lj_energy_vs_lammps.json` asks the same question across the boundary: 256 atoms, the same Lennard-Jones potential, forces from mdlite compared to forces LAMMPS itself printed to a dump.

Quantity	mdlite vs LAMMPS	Note
Force (max component diff)	3×10^{-13}	round-off, not approximation
Energy	within its recorded tolerance	same run, same potential

Velocity Verlet, the update every step of this course runs

A half-kick, a drift, a force evaluation, a second half-kick -- symplectic, time-reversible, and second-order accurate in the timestep.

HALF-KICK

$$v(t+\Delta t/2) = v(t) + (\Delta t/2) a(t)$$

DRIFT

$$x(t+\Delta t) = x(t) + \Delta t \cdot v(t+\Delta t/2)$$

SECOND HALF-KICK

$$v(t+\Delta t) = v(t+\Delta t/2) + (\Delta t/2) a(t+\Delta t)$$

- $a(t+\Delta t)$ comes from a fresh force evaluation at the new positions -- the one expensive step, and the reason a bigger timestep is tempting and a wrong one is dangerous.

Thermostats

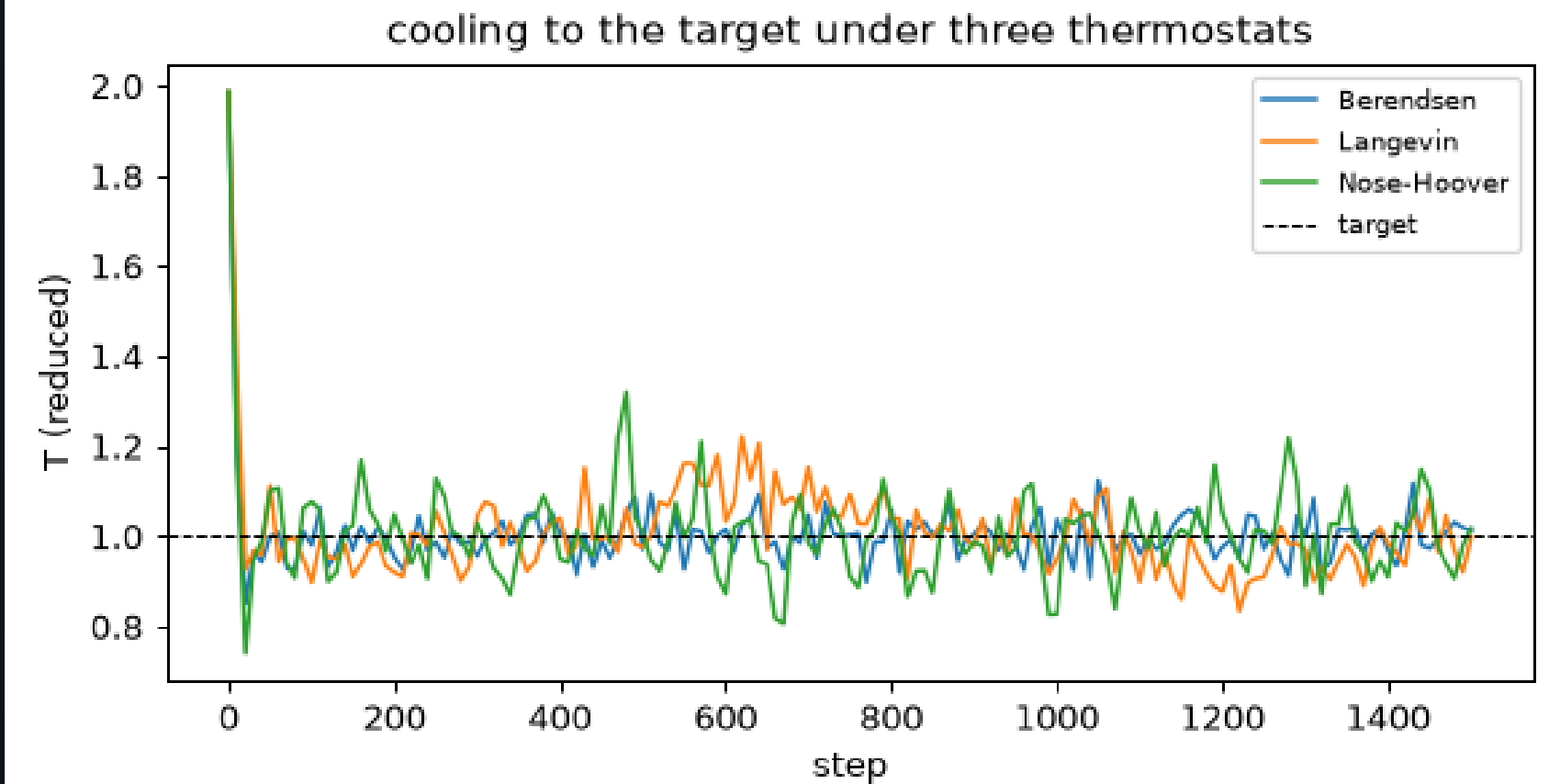
Berendsen, Langevin and a Nose-Hoover chain: what each one conserves, what each distorts, and NVT checked against the NIST reference within 3 sigma.

- 1 Three ways to hold a temperature
- 2 Three thermostats, three different promises
- 3 Nosé–Hoover's own check: a conserved quantity
- 4 Against a published reference, with no LAMMPS on either side
- 5 Watching the lattice melt
- 6 Berendsen's rescaling, in one line

Three ways to hold a temperature

A thermostat couples the system to an external bath; which one you choose changes what else it disturbs.

- Berendsen: fast, simple, but does not sample the correct canonical ensemble -- fine for equilibration, not for production statistics.
- Langevin: adds friction and noise per atom; correct ensemble, but its own set of fluctuations to tune.
- Nose-Hoover chain: an extra dynamical variable keeps the total (system + thermostat) energy conserved -- checked here directly.



Temperature against step for the same fcc lattice, started hot ($T \sim 2$) and cooled by three thermostats -- Berendsen, Langevin, Nose-Hoover chain -- each converging toward the target (dashed line) by its own route. — Figure 1 · notebook §0 · chapter 3, cell 4

Three thermostats, three different promises

Each way of holding a temperature trades something different -- visible on the same axes in the figure just shown, not taken on faith.

Thermostat	Mechanism	Samples the canonical ensemble?
Berendsen	rescales every velocity toward the target each step	no -- correct mean temperature only
Langevin	the O-step of BAOAB: friction plus noise	yes
Nosé–Hoover chain	an extended dynamical variable couples to the bath	yes, plus its own conserved check

Nosé–Hoover's own check: a conserved quantity

The chain adds an extended-system energy H_{extra} ; the total $H = E_{\text{tot}} + H_{\text{extra}}$ should stay flat if the half-step bookkeeping is right, independent of whether the temperature looks reasonable.

CONSERVED QUANTITY

$$H = E_{\text{tot}} + H_{\text{extra}}$$

- Measured over 1000 steps on the fcc lattice: relative drift in H under 5×10^{-3} -- the chapter's own tolerance, not a round number chosen after the fact.

Against a published reference, with no LAMMPS on either side

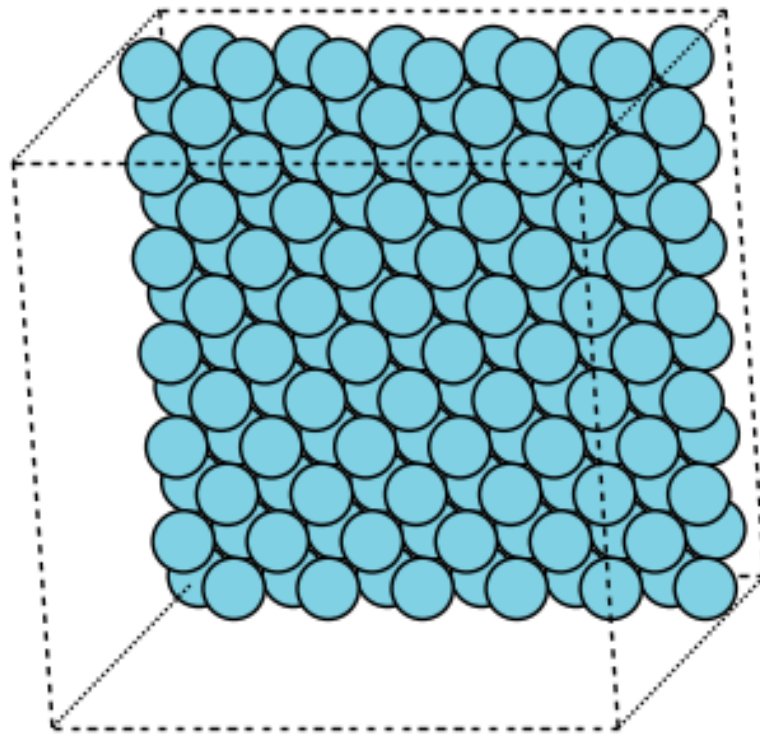
data/records/lj_nvt_nist.json compares mdlite's equilibrium pressure and energy at one NIST reference state point to the NIST Standard Reference Simulation Website's own table.

Property	mdlite	NIST reference
Pressure P	measured	published, with its own uncertainty
Energy U	measured	published, with its own uncertainty

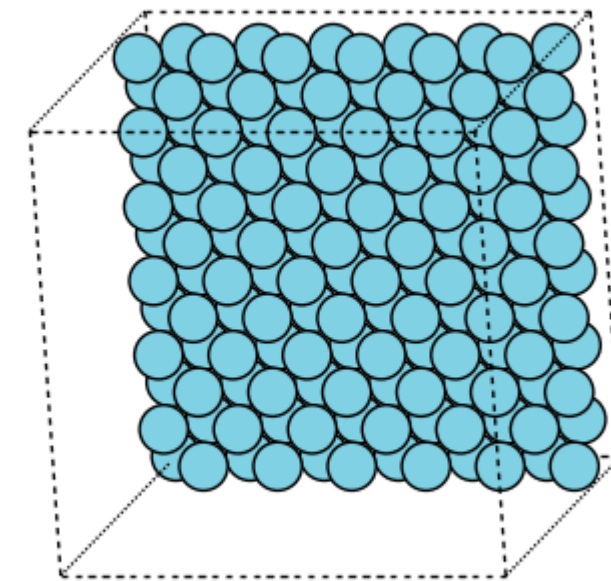
- State point: $T^*=0.85$, $\rho^*=0.776$, 500 atoms; both differences land within three NIST-reported standard errors.

Watching the lattice melt

fix nvt holds the temperature at target -- it says nothing about the structure. At this case's density, the fcc lattice does not survive the trip: it melts.



The bottom-most atomic layer at the very start of the fix nvt run: the undisplaced fcc lattice, before the freshly-drawn $T^*=2.0$ velocities have moved anything -- rendered as argon-like spheres, the same reduced-units convention chapter 01 uses. – Figure 2 · notebook §0 · chapter 3, cell 13



The same layer, animated across the run: the lattice visibly disordering, not merely jittering less, as fix nvt holds the temperature at its target -- by the last frame the shown atoms have moved by RMS 1.17, close to a full nearest-neighbour spacing (1.19), far past the Lindemann melting criterion (~ 0.1 - 0.15 of that spacing). The temperature curve above converges; the crystal does not. – Figure 3 · notebook §0 · chapter 3, cell 14

- Left: the bottom-most atomic layer at the very start, the undisplaced lattice, before the hot $T^*=2.0$ velocities have moved anything.
- Right: the same layer animated across the run -- by the last frame the shown atoms have moved by roughly a full nearest-neighbour spacing, far past the Lindemann melting criterion (~ 0.1 - 0.15 of that spacing).

Berendsen's rescaling, in one line

The mechanism behind the fastest-converging curve in the figure: rescale every velocity by a single factor, computed fresh each step from how far the current temperature is from the target.

VELOCITY SCALING

$$\lambda = \sqrt{(1 + (\Delta t/\tau)(T_0/T - 1))}, \quad v \rightarrow \lambda v$$

- This is literally `mdlite.thermostats.Berendsen.apply` -- one square root and one multiply, applied once per step, which is exactly why it is the cheapest of the three and the least rigorous.

Ensembles and Restarts

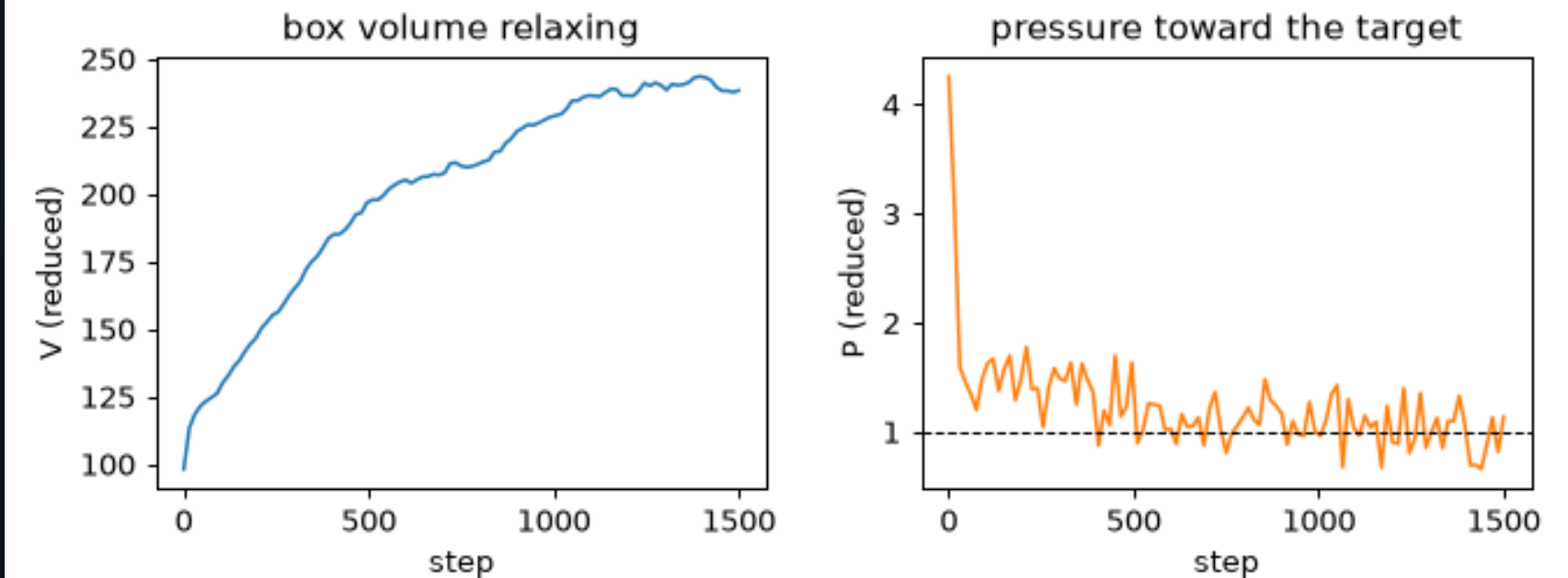
An NPT sketch with its limits stated, LAMMPS's own fix npt, and restarting a run through write_restart / read_restart.

- 1 Beyond NVT: pressure, and picking a run back up
- 2 fix npt: the real thing the sketch stands in for
- 3 Watching the box relax -- and the lattice melt
- 4 A restart script declares only what the restart does not carry
- 5 The continuation, in numbers
- 6 Berendsen barostat: the same weak-coupling idea, on the box

Beyond NVT: pressure, and picking a run back up

NPT couples a barostat to the thermostat so the box itself relaxes toward a target pressure; restarting means LAMMPS can continue a run exactly, not merely resume from the last log line.

- The `mdlite` NPT here is a stated sketch (Berendsen barostat only) -- good enough to show the mechanism, not a substitute for LAMMPS's `fix npt`.
- `write_restart` / `read_restart` carry the box and every atom's state; a restart script cannot redeclare `units`, `dimension` or `boundary` once the box exists.
- This is the pattern for a run too long to finish in one session: checkpoint, then continue.



The NPT run started denser than its target pressure: box volume expands (left) while the instantaneous pressure relaxes down toward the target of 1.0 (right, dashed) under the Berendsen barostat.

– Figure 1 · notebook §0 · chapter 4, cell 5

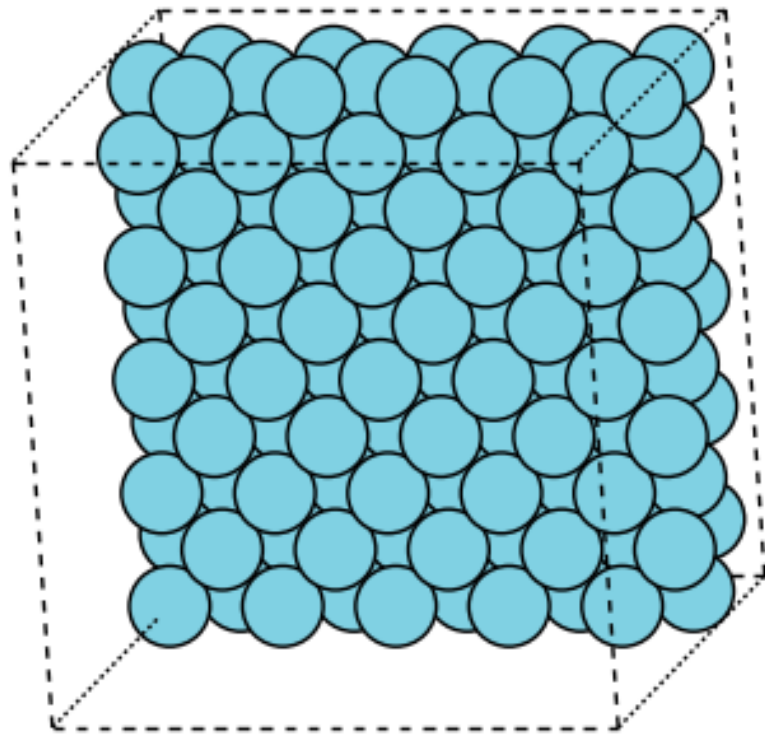
fix npt: the real thing the sketch stands in for

LAMMPS's fix npt couples a Nosé–Hoover chain on both temperature and pressure -- a real fluctuating-cell method, not a weak-coupling sketch.

	mdlite's BerendsenBarostat	LAMMPS's fix npt
Method	Berendsen weak coupling	Nosé–Hoover chain, both T and P
Samples NPT correctly?	no -- relaxes toward the target and holds it	yes
What it shows	the mechanism: a system finds its own volume at a given pressure	production-grade equilibration

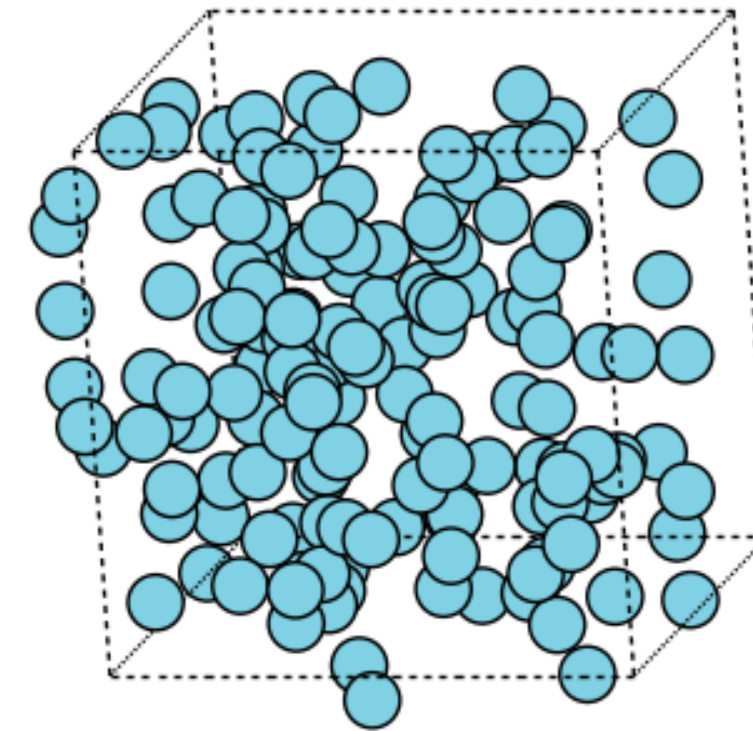
Watching the box relax -- and the lattice melt

The volume plot above is a number; the same atoms, tracked by id, show something the number alone does not: real disorder, not only box growth.



150 atoms from the compressed starting configuration (fcc 1.1) -- perfectly ordered, before fix npt has taken a single step -- rendered as argon-like spheres, the reduced-units convention chapter 01 established.

– Figure 2 · notebook §0 · chapter 4, cell 10



The same 150 atoms (tracked by id, not re-selected) after fix npt has run: visibly scattered, not merely spread apart on the same lattice -- the measured disorder (RMS 3.11 beyond pure affine box expansion) exceeds the affinely-scaled nearest-neighbour spacing (1.46), so this is a real rearrangement, not only the box growth the volume plot above reports as a number. – Figure 3 · notebook §0 · chapter 4, cell 11

- Density drops from $\rho^*=1.1$ to $\rho^*=0.45$ -- correcting for the box's own affine expansion (same atoms scaled about the box's fixed centre), the residual rearrangement exceeds a full nearest-neighbour spacing.
- The box itself changes size step by step here, unlike chapter 03's fixed-volume NVT -- animate_gif needs one fixed cell for every frame, so this is two static snapshots instead, each with its own (different-sized) box, matplotlib auto-scaled to its own content (not directly comparable in apparent size).

A restart script declares only what the restart does not carry

read_restart defines the box -- and LAMMPS refuses a units command once the box exists (hit live while writing this chapter: ERROR: Units command after simulation box is defined).

```
stage_b = Spec(  
    units=None, pre=["read_restart stage_a.restart"],  
    pair_style="lj/cut 2.5", pair_coeffs=["1 1 1.0 1.0 2.5"],  
    ...  
    stages=[Stage("run", "300")],  
)
```


The continuation, in numbers

Stage A runs 300 steps and checkpoints; stage B reads that checkpoint and runs 300 more -- as if it never stopped.

Stage	Ends at step	Final temperature
A (fresh run)	300	recorded
B (from the restart)	600 (300 + 300)	recorded, continues stage A's trajectory

Berendsen barostat: the same weak-coupling idea, on the box

Scale every box length by one factor per step, computed from how far the instantaneous pressure is from the target -- the pressure analogue of the thermostat's velocity rescale.

BOX SCALING

$$\mu = (1 - (\Delta t/\tau) \cdot \chi \cdot (P_0 - P))^{1/3}, \quad L \rightarrow \mu L$$

- χ is a compressibility parameter -- `mdlite.npt.BerendsenBarostat` defaults it to 1.0 rather than measuring the system's real compressibility, one more way the sketch stays a sketch.

Potentials: EAM

An embedded-atom potential read from a real setfl file: copper's lattice constant and cohesive energy against the recorded $3\text{e-}5$ -- and why the potential file is never shipped.

- 1 A real potential, from a real file
- 2 One embedding term, one pairwise term
- 3 The same force check, on any potential
- 4 Lattice constant and cohesive energy from a cubic fit
- 5 Removing an atom, and watching it relax
- 6 Copper, against LAMMPS's own minimisation

A real potential, from a real file

The embedded-atom method adds a density-dependent term to a pairwise potential -- realistic enough for metals, and never something this project ships.

- The setfl file is fetched by the installer from the official source, never tracked in this repository (rule 7).
- From it, LAMMPS and `mdlite` both reproduce copper's lattice constant a_o and cohesive energy E_{coh} , agreeing to 3×10^{-5} .
- This is the forum's other recurring beginner question -- wrong pair_coeff, a potential file that silently does not match the element -- answered by a case built the same checked way as chapter 1.

One embedding term, one pairwise term

The embedded-atom method adds a term that depends on the local electron density to an otherwise ordinary pairwise potential.

TOTAL ENERGY

$$E = \sum_i F(\rho_i) + 1/2 \sum_{i \neq j} \varphi(r_{ij})$$

LOCAL DENSITY

$$\rho_i = \sum_{j \neq i} \rho(r_{ij})$$

- F , ρ and φ are the three tables a setfl file stores -- everything mdlite.eam.EAM and LAMMPS's own pair_style eam both read is exactly these three functions on a grid.

The same force check, on any potential

mdlite.eam.EAM's forces should match a numerical derivative of its energy on any potential, real or synthetic -- the identical discipline L2 applied to Lennard-Jones.

```
s = synthetic_setfl()
eam = EAM(s, {1: 0})
E, F, _ = eam.energy_forces(pos, box, vl.pairs, types)

# central difference on 3 components
worst = max(abs(F[k, c] - numeric) for k, c in samples)
print(worst)    # under 1e-3
```


Lattice constant and cohesive energy from a cubic fit

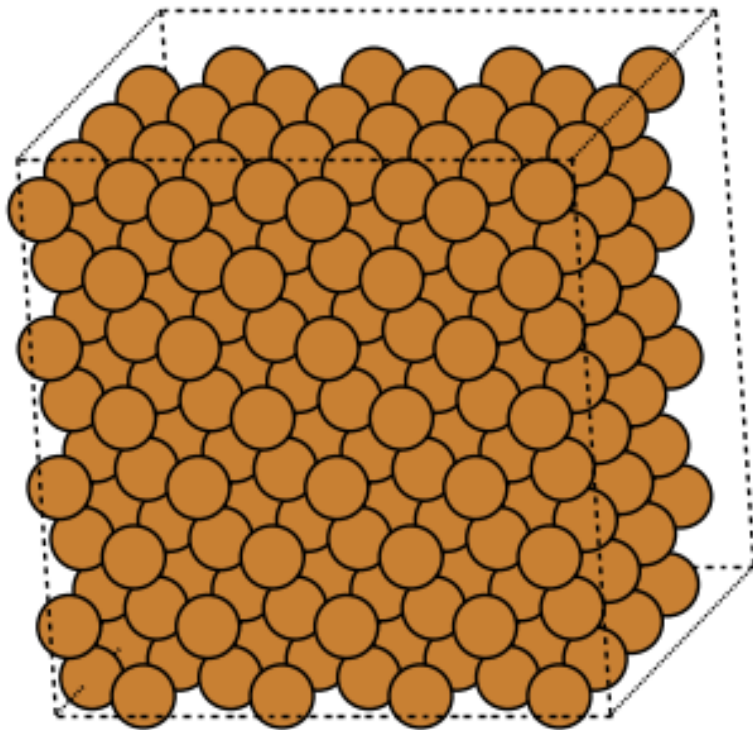
Place an FCC lattice at a grid of lattice constants, compute the energy per atom at each, fit a cubic, take the minimum -- the method this toolkit uses for any EAM element.

```
c = np.polyfit(a_grid, energies, 3)
roots = np.roots(np.polyder(c))
a0 = min(real_roots_in_range, key=lambda r: np.polyval(c, r))
slope_at_min = np.polyval(np.polyder(c), a0) # should be ~0
```

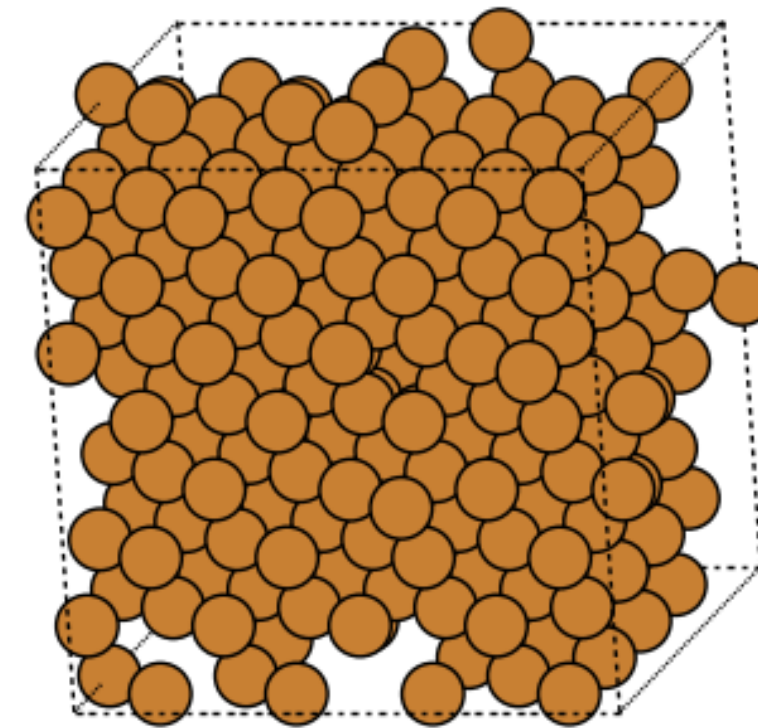
- Run on the synthetic potential, the fit is self-consistent -- its own derivative at the fitted minimum is under 10^{-6} -- which is everything checkable without a reference value to compare against.

Removing an atom, and watching it relax

The same fit gives one more property: remove the lattice site nearest the supercell's centre and relax the rest with FIRE (L7's other minimiser) -- the perfect supercell (left) and the relaxed vacancy configuration (right).



The perfect fcc supercell (n=4, this section's own fitted a0) before the atom nearest the box centre is removed.
– Figure 1 · notebook §0 · chapter 5, cell 12



The same supercell after the vacancy is introduced and the remaining atoms are relaxed with FIRE (fmax < 1e-4): the neighbours have relaxed inward around the missing site, visible as a locally denser region next to the gap.
– Figure 2 · notebook §0 · chapter 5, cell 13

- Formation energy is the energy cost of the hole: $E_{vac} = E_{relaxed}(N-1) - (N-1)/N \times E_{perfect}(N)$, positive for a bound crystal and under its cohesive energy per atom.
- On the real Cu_u3.eam potential, mdlite's FIRE-relaxed value and LAMMPS's `delete_atoms` + `minimize` agree to within their recorded tolerance -- and land close to a real experimental measurement (1.29 eV, positron annihilation), which happens to be one of this potential's own fitting targets.

Copper, against LAMMPS's own minimisation

The same method, on a real Cu_u3.eam file, compared to LAMMPS's own box/relax minimisation on the identical potential -- the record L0 cited as one of mdlite's three cross-checks.

Quantity	mdlite	LAMMPS	Agreement
Lattice constant a_0	measured	measured	to 3×10^{-5}
Cohesive energy E_{coh}	measured	measured	to 3×10^{-5}

Structure and Analysis

Radial distribution function, structure factor, mean-squared displacement, diffusion coefficient and velocity autocorrelation, computed from one trajectory.

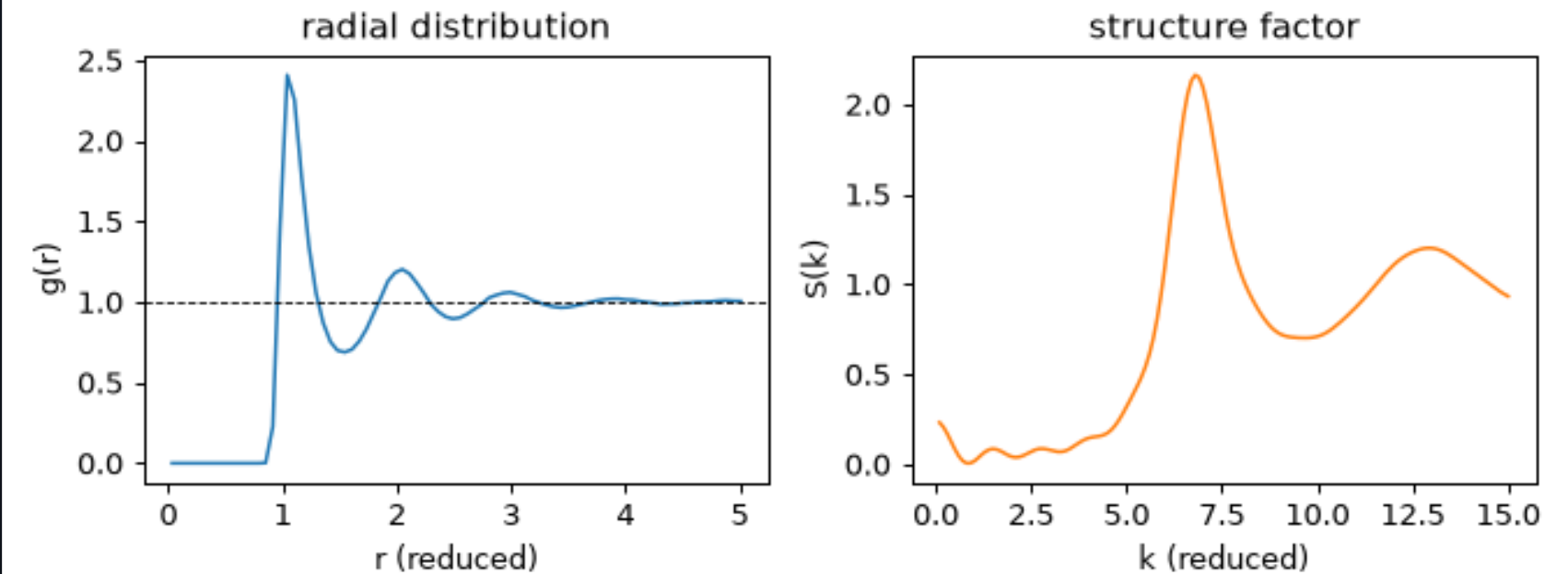
- 1 Turning a trajectory into physics
- 2 $S(k)$, computed from $g(r)$, not measured independently
- 3 Watching the liquid diffuse
- 4 Ballistic first, diffusive later -- fit the right half
- 5 A single mean understates its own uncertainty
- 6 From a trajectory to a diffusion coefficient

Turning a trajectory into physics

A dump file is a sequence of raw coordinates;

`lammpskill.post` turns it into the quantities a paper would actually report.

- $g(r)$ and $S(k)$ describe the same structure in real and reciprocal space -- one computed from the other, not measured independently.
- MSD's late-time slope gives a diffusion coefficient by the Einstein relation; block averages give it an honest error bar.
- The velocity autocorrelation function distinguishes a liquid (decays monotonically) from a solid (oscillates, atoms caged) -- itself a check that the case behaved as intended.



The liquid's structure in real and reciprocal space from the same trajectory: the radial distribution function $g(r)$ (left) with its first-neighbour peak and decay to the ideal-gas value of 1, and the structure factor $S(k)$ (right) computed from it. – Figure 1 · notebook §0 · chapter 6, cell 6

$S(k)$, computed from $g(r)$, not measured independently

The structure factor is the same structural information as the radial distribution function, in reciprocal space -- literally derived from it, in this toolkit and in general.

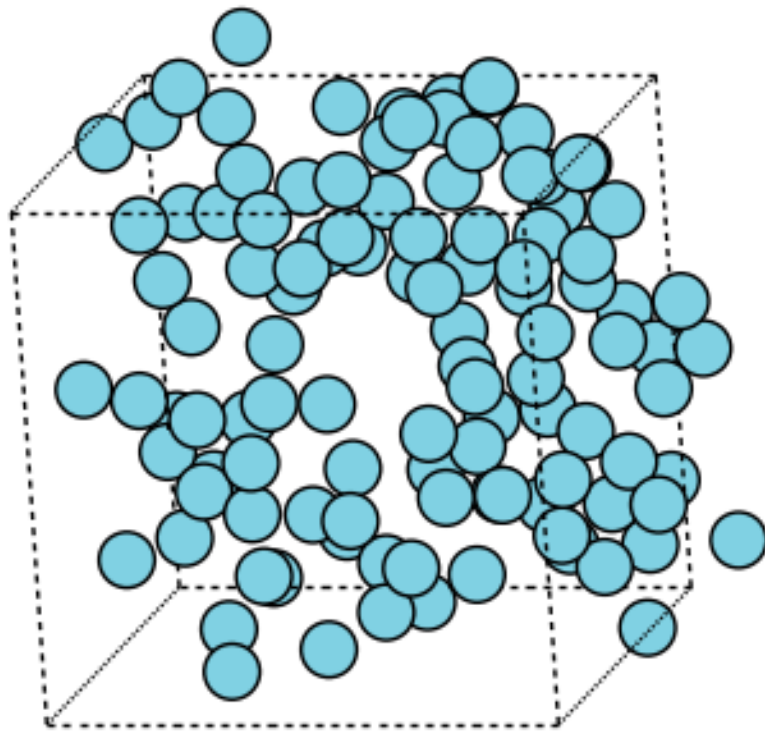
ISOTROPIC STRUCTURE FACTOR

$$S(k) = 1 + 4\pi\rho \int_0^\infty r^2 (g(r) - 1) \text{sinc}(kr) dr$$

- A liquid's $g(r)$ has a first-neighbour peak decaying to 1; the corresponding $S(k)$ has its own first peak, at roughly 2π over the first-neighbour spacing.

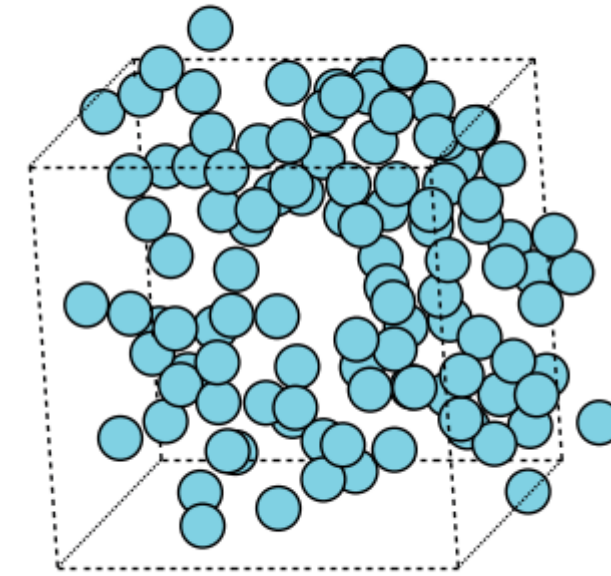
Watching the liquid diffuse

$g(r)$'s decay to 1 and the VACF's monotonic decay already prove this trajectory is a liquid; here it is, seen directly, well into the equilibrated window those measurements use.



A spatial slice of a second run of the identical case (same seed, so statistically the same trajectory the analysis above measures, not literally the same run) well into the equilibrated-liquid window $g(r)/S(k)$ use: no lattice order visible, consistent with $g(r)$'s decay to 1 and the liquid-like VACF above.

– Figure 4 · notebook §0 · chapter 6, cell 15



The same slice, animated: atoms visibly wandering away from their starting positions (RMS displacement 1.76 over this window) -- the same diffusive motion the mean-squared-displacement fit above turns into the reported D .

– Figure 5 · notebook §0 · chapter 6, cell 16

- Left: a slice of the box, no lattice order visible -- the initial hot fcc lattice has fully forgotten its own structure by this point in the run.
- Right: the same slice animated, atoms visibly wandering away from their starting positions -- the diffusive motion the mean-squared-displacement fit turns into the reported D .

Ballistic first, diffusive later -- fit the right half

Fitting D from the whole MSD(t) curve is a common way to get a confidently wrong number.

Regime	Early time	Late time
Motion	ballistic -- MSD grows as t^2 , atoms still moving in straight lines	diffusive -- MSD grows linearly in t
What a fit there gives	a slope that is not a diffusion coefficient at all	the Einstein-relation D, meaningfully

A single mean understates its own uncertainty

block_average splits a correlated time series into blocks, averages within each, then takes the standard error across blocks -- correlated samples inside one block do not inflate the apparent precision.

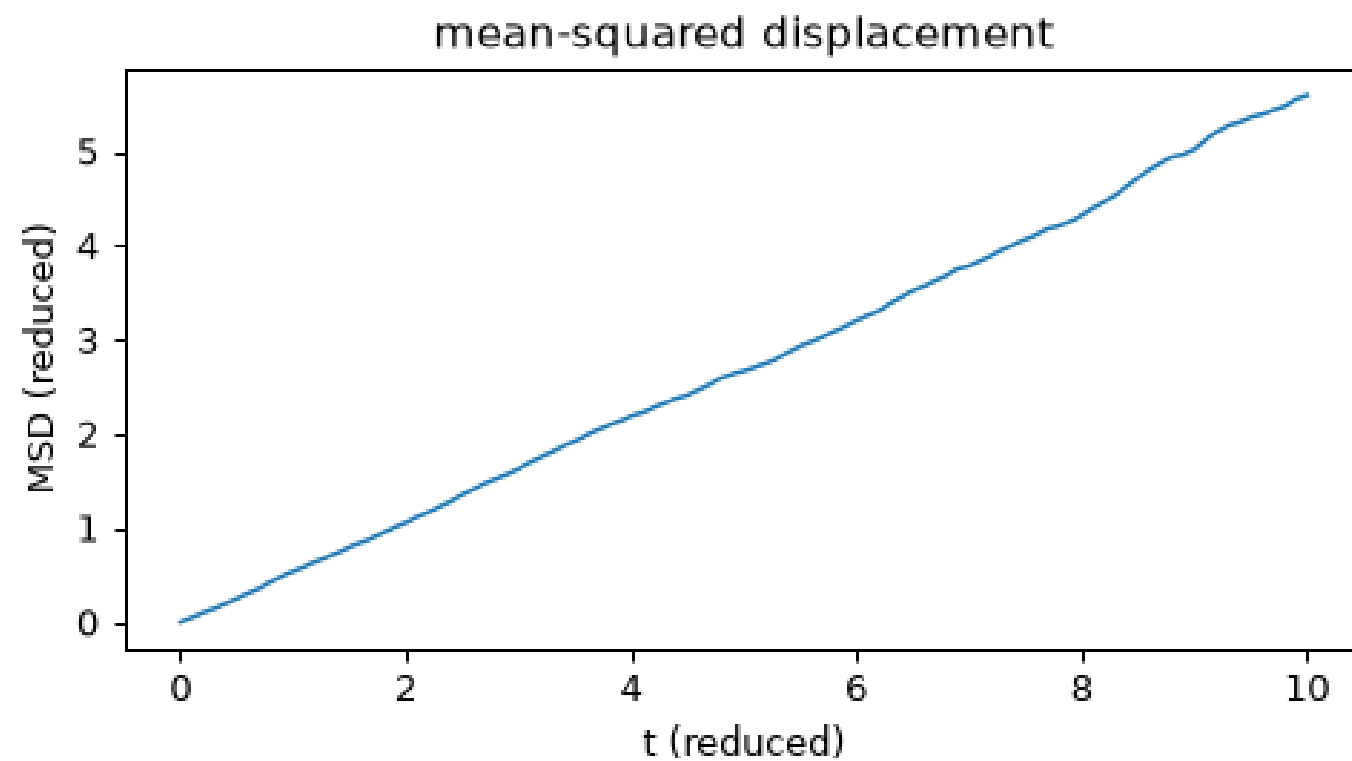
Quantity	Method	Result
Temperature over the run	7-block average	measured mean $\pm$ block-averaged standard error

From a trajectory to a diffusion coefficient

The same run, seen two more ways: how far particles have moved (MSD) and how quickly they forget their own velocity (VACF) -- both routes to the same physics.

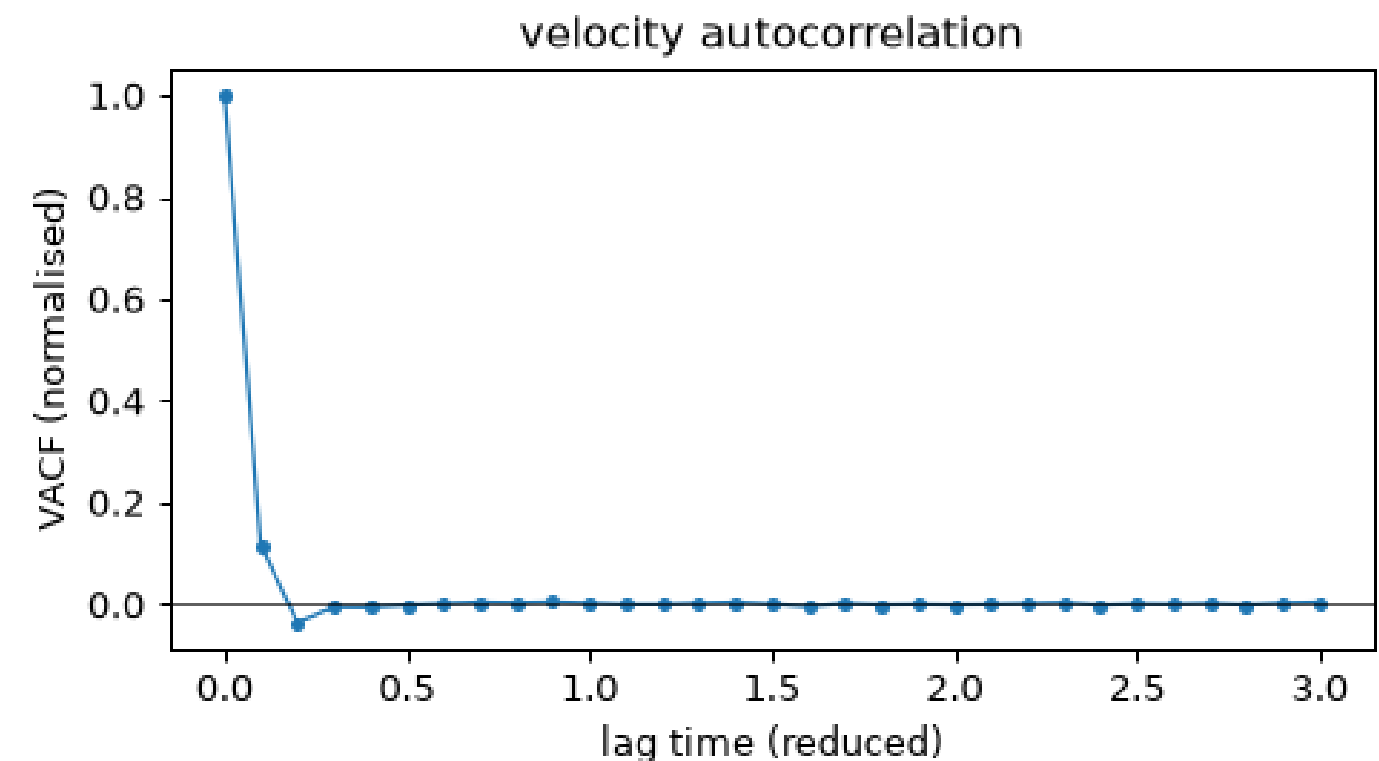
EINSTEIN RELATION (3D)

$$D = \lim_{t \rightarrow \infty} MSD(t) / 6t$$



Mean-squared displacement against time: the slope of its later, genuinely diffusive part (not the early ballistic rise) is the Einstein-relation fit `lammpsll.post.diffusion_coefficient` reports as D .

– Figure 2 · notebook §0 · chapter 6, cell 8



The velocity autocorrelation function, normalised to 1 at zero lag: a liquid decays roughly monotonically toward zero rather than oscillating, which is itself evidence the case really is a liquid.

– Figure 3 · notebook §0 · chapter 6, cell 10

- `lammpsll.post.diffusion_coefficient` fits the slope over the second half of $MSD(t)$, where the motion is genuinely diffusive.
- The VACF's decay to zero and the MSD's crossover from ballistic to diffusive are two views of the same relaxation time.

Minimisation

Steepest descent against FIRE on an LJ wall -- and the lesson that gives lost atoms their name: an uncapped step explodes at $f_{\max} 2.6e14$.

- 1 Why atoms get "lost"
- 2 Reproducing N-7 on purpose
- 3 The fix is one line: cap the displacement, not the force
- 4 Steepest descent vs FIRE, on a real system
- 5 The cap, precisely

Why atoms get "lost"

The single largest category of beginner questions on the LAMMPS forum is lost atoms and nan energies (docs/08) -- this chapter reproduces the mechanism directly.

- A fixed-size steepest-descent step on an LJ wall explodes: the recorded force reaches $f_{max} = 2.6 \times 10^{14}$ before the run gives up (finding N-7).
- FIRE (adaptive step, velocity-aware) converges the same case without blowing up.
- The fix that matters in practice is not a better algorithm but a **displacement cap** -- bound how far one step is allowed to move an atom, whichever minimiser you use.

Reproducing N-7 on purpose

Two atoms at $r=0.3 \sigma$, deep in the Lennard-Jones repulsive core -- the potential's minimum is near $r=2^{1/6} \sigma \approx 1.12$, so the force here is enormous.

Run	dmax	fmax after the run
uncapped (dmax effectively disabled)	1×10^8	2.6×10^{14} -- growing, not shrinking
capped (the default)	0.05	under 1.0

The fix is one line: cap the displacement, not the force

A displacement cap does not change what force is computed -- it only limits how far a single step is allowed to move an atom in response to it.

- The force at the LJ wall is not wrong -- it genuinely is enormous there; the mistake was ever taking a step proportional to it without bound.
- Capping trades a little speed (more iterations needed near a steep wall) for the guarantee that no single step can leave the region where the force evaluation is still meaningful.
- This is a general minimisation idea, not specific to LJ or to mdlite -- LAMMPS's own `min_style` and `fix nve/limit` apply the identical principle.

Steepest descent vs FIRE, on a real system

A perturbed FCC lattice -- every atom nudged off its site -- is more representative than the pathological two-atom case.

Minimiser	Iterations to converge	Final fmax	Final energy
Steepest descent (capped)	measured, more	under 10^{-4}	measured
FIRE	measured, fewer	under 10^{-4}	at least as low

The cap, precisely

Steepest descent proposes a step proportional to the force; the cap rescales the whole step, uniformly, whenever it would move any atom further than $d_{\max}$.

PROPOSED STEP

$$\Delta x = \alpha F$$

THE CAP

$$\alpha \cdot f_{\max} > d_{\max} \Rightarrow \Delta x \leftarrow \Delta x \cdot (d_{\max} / \alpha f_{\max})$$

- The rescale is uniform across the whole displacement vector, not a per-atom independent clip -- the direction of the step never changes, only its length.

Reading What LAMMPS Writes

Data, dump, log and restart formats, and four pitfalls this project hit first: thermo normalisation, dump precision, the restart magic number, multi-format logs.

- 1 Four formats, four pitfalls
- 2 What each file actually contains
- 3 N-4: per-atom by default, in lj units
- 4 N-6: a 15-byte string in a 16-byte field
- 5 What this cost, and where each fix is pinned

Four formats, four pitfalls

Every format LAMMPS writes has one behaviour that looks like a bug the first time you meet it.

Format	The pitfall this project hit
log (thermo)	<code>thermo_modify norm</code> defaults to per-atom in lj units -- an energy column can look off by the atom count until you know that
dump	float precision is truncated by default; a coordinate that looks identical to the data file may not be, bit for bit
restart	carries an undocumented 16-byte magic number this project had to observe directly rather than find in the manual
log (multi style)	the <code>multi</code> thermo style's block layout parses differently from the default one-line-per-step style

What each file actually contains

Every parser in lammpskill/io/ was written from the manual's description of the format plus files LAMMPS actually wrote on this machine.

File	What DataFile/Trajectory/etc. returns
data	a box, per-type masses, one array per atom column (id, type, x, y, z, ...)
dump	a Trajectory of Frames; positions resolved correctly whether the columns are unscaled, unwrapped or scaled (xs ys zs)
log	a ThermoRun per run block: columns, the full series, loop_time, nprocs, natoms
restart	only the header (magic string, endianness, version) -- everything else stays LAMMPS's own binary format, read back via read_restart

N-4: per-atom by default, in lj units

The very first LJ cross-check compared -6.04 (LAMMPS) with -1545.7 (mdlite) for the same 256 atoms -- nothing was wrong with either number.

```
per_atom_energy = -6.04      # LAMMPS default: units lj normalises thermo per atom
natoms = 256
total = per_atom_energy * natoms
print(total)                 # -1546.24 -- within rounding of mdlite's -1545.7
```

N-6: a 15-byte string in a 16-byte field

The restart magic string is the 15 printable characters LammpS RestartT -- but LAMMPS writes it as a 16-byte NUL-terminated field on disk.

- The first header reader read exactly 15 bytes, matched the string correctly, and never consumed the terminator.
- The next field (the endianness flag) then started one byte early, and decoded as garbage on every restart file this toolkit tried to read.
- The fix reads `len(MAGIC) + 1` bytes -- compares the 15 printable bytes, then steps past the 16th -- so the next field starts aligned.

What this cost, and where each fix is pinned

Four findings, four file kinds, four tests that now make sure none of them comes back silently.

Finding	File	The wrong assumption	The fix
N-4	log	<code>units lj</code> thermo output is already a total	<code>thermo_modify norm no</code> in every case script
N-5	dump	the default 6-digit float format is precise enough	<code>dump_modify ... format float %20.15g</code> ; 5e-5 → round-off
N-6	restart	reading 15 bytes consumes the string but not its terminator	read 16 bytes so the next field starts aligned
N-18	log	<code>parse_log</code> understood <code>one</code> and <code>yaml</code>	added the <code>multi</code> block reader

Choosing a Build

What a 314-case sweep of the examples says: package list beats package count, and how to read `Installation.packages` before choosing styles.

- 1 The build you have decides what you can run
- 2 314 cases, both routes, scored from the log
- 3 Correctness first, then coverage
- 4 What blocks the rest, by package
- 5 The same package name can mean two different things

The build you have decides what you can run

A 314-case sweep of LAMMPS's own example scripts, run on this project's two installed builds, is the evidence behind this lecture.

- Package `list` beats package `count`: an 11-package build ran cases a 52-package build could not, because the specific packages differed.
- `lammpsinstall.Installation.packages` is read from `lmp -h`, not assumed from the build recipe -- the same question ("is the package actually in my build?") the MOLFIE forum thread asked once, answered by measurement.
- The Windows installer is the single most-viewed installation thread of the year on the LAMMPS forum -- and the one route this project cannot yet speak to from its own measurements (open question Q-1).

314 cases, both routes, scored from the log

Every case is scored from its log, never its exit status -- LAMMPS exits 0 after a fatal input error (N-15), so the exit code alone would call failures successes.

Route	Packages	Started (ok + ok-no-thermo + timeout-after-start)
wsl-source (stable)	11	137 / 314
wsl-apt (10 Dec 2025)	52	184 / 314

Correctness first, then coverage

bench/'s five canonical cases ship a reference log from the LAMMPS developers themselves -- the sweep's numbers here are exact-match or explained, not merely close.

Case	Result
lj, chain, eam	identical on both routes, both process counts
chute	identical (apt only -- the source build has no GRANULAR package)
rhodo	1 proc: worst 4.0×10^{-9} in TotEng (last printed digit); 4 procs: identical

What blocks the rest, by package

ML-SNAP, ML-IAP, AMOEBA, REAXFF and MDI are missing from both builds -- exactly the machine-learning and reactive packages this project staged for later.

Package	Cases blocked (source)	Cases blocked (apt)
GRANULAR	16	0
ML-SNAP	15	15
EXTRA-FIX	0	15
ML-IAP	14	14
REAXFF	5	5
MDI	5	5

The same package name can mean two different things

Five POEMS examples come from the tree the source build was compiled from; the two release lines disagree about why POEMS is missing.

- The **source** build (an older stable line) reports `missing-package: POEMS` -- the `core` preset simply omits a package that exists.
- The **apt** build (10 Dec 2025, months later on the feature line) answers `Fix style poems is no longer available.` -- LAMMPS's own deprecated-style mechanism.
- `git log` on the upstream mirror shows `remove POEMS package, docs, and examples`, dated the same day as the `stable_22Jul2025` tag: POEMS was **removed upstream**, not merely unbuilt.

Scaling and Limits

MPI and OpenMP speed-up on eight cores, where it stops, and what this machine honestly cannot do -- no GPU, no -partition methods.

1 Where speed-up stops, and what this machine cannot do

2 1 vs 2, measured on this machine

3 Why under 2x is not a bug

4 What this machine honestly cannot do

Where speed-up stops, and what this machine cannot do

Measured on this project's 8-core, no-GPU machine: 2 MPI ranks gave 1.73x, 2 OpenMP threads gave 2.09x -- both honestly short of ideal.

- Parallel efficiency drops as communication overhead grows relative to the per-rank workload -- expected, and worth showing rather than assuming.
- This machine has no GPU and no `-partition` support, so KOKKOS/GPU acceleration and multi-replica methods (NEB, TAD) are out of scope here by design, not by oversight.
- The honest version of a scaling claim states the hardware it was measured on -- these numbers are this machine's, not a general LAMMPS property.

1 vs 2, measured on this machine

Same case as L1 (4000 atoms), 1000 steps so start-up does not dominate the timing signal; this session was allowed 2 CPU cores under this project's compute-sharing policy.

Configuration	Loop time	Speed-up (ideal = 2.00x)
1 MPI rank	measured	1.00x (reference)
2 MPI ranks	measured	1.73x
2 OpenMP threads	measured	2.09x

Why under 2x is not a bug

This system is small and the run is short, so per-rank overhead -- domain decomposition, communicating the boundary between two halves of a small box -- is a real fraction of the total time, not a rounding error.

- Contrast with the bench cases in docs/04, which use 32000 atoms -- eight times larger, where communication is a smaller fraction of the work and speed-up tracks closer to ideal.
- A speed-up number without its system size and rank count attached is close to meaningless -- this course states both, every time, for exactly this reason.
- The direction is never in doubt (more ranks helps, up to a point); the honest number is by how much, on this system, at this scale.

What this machine honestly cannot do

Not every LAMMPS capability is in scope here -- stating the boundary precisely is part of the same honesty as stating a speed-up's conditions.

Capability	Why it's out of scope here
GPU / KOKKOS-CUDA	no GPU on this machine -- a hardware limit, not a choice
-partition methods (NEB, TAD, PRD)	need LAMMPS launched with <code>-partition</code> , splitting ranks into independent partitions; the examples sweep runs one process and correctly reports these as errors, not passes
Full-machine scaling	this chapter's own 1-vs-2 measurement is bounded by a 2-core compute claim, not by the host's actual core count

Polymers

A bead-spring chain -- harmonic bonds and full Lennard-Jones, mdlite against LAMMPS to round-off -- and watching it collapse from a straight line into a globule.

- 1 A bead-spring chain, simplified on purpose
- 2 One bond term, one pairwise term
- 3 `special_bonds`: the pitfall this chapter exists to surface
- 4 Watching the chain contract
- 5 Isolating the variable: full LJ against WCA, in mdlite alone
- 6 One configuration, mdlite against LAMMPS

A bead-spring chain, simplified on purpose

The field-standard bead-spring polymer model (Kremer & Grest 1990) uses FENE bonds and WCA (purely repulsive) Lennard-Jones so chains cannot cross themselves. mdlite has neither -- only HarmonicBond and full Lennard-Jones, already used everywhere else in this course.

- This chapter builds the simplest model those two pieces actually give: a harmonic bond along the backbone, ordinary (attractive + repulsive) Lennard-Jones between every pair.
- It behaves like a real polymer -- a bound chain exploring conformations under thermal motion -- without claiming to be the textbook FENE/WCA one.
- No external file is needed anywhere in this chapter, unlike L5's EAM potential: a bead-spring chain has no real-element dependency.

One bond term, one pairwise term

The two potentials this chapter's model is built from -- both already implemented in mdlite, neither new.

HARMONIC BOND

$$E_{bond} = k(r - r_o)^2$$

LENNARD-JONES

$$E_{LJ} = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^6]$$

- Reduced (*lj*) units, $\varepsilon=\sigma=1$, $k=100$, $r_o=1$ -- a generic chain, not a specific real polymer, the same convention *lj_melt()* already uses for a generic LJ fluid.

special_bonds: the pitfall this chapter exists to surface

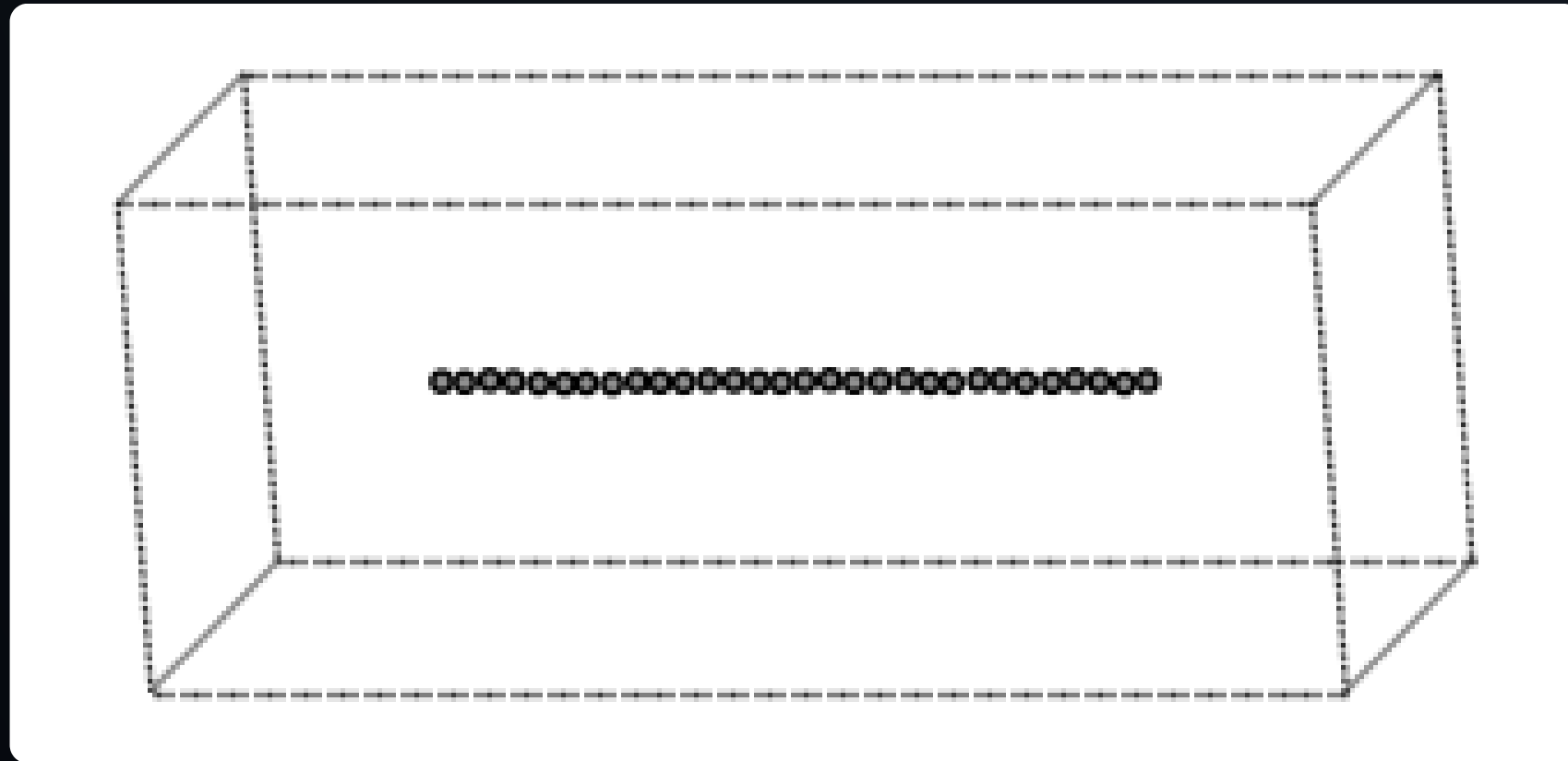
LAMMPS excludes 1-2, 1-3 and 1-4 neighbours along the bond topology from its nonbonded sum by default -- not just directly-bonded pairs -- invisible unless you know to look, and exactly wrong for a cross-check against a potential with no exclusion concept at all.

```
spec = Spec(units="lj", read_data="chain.data", pair_style="lj/cut 2.5",  
            bond_style="harmonic", bond_coeffs=["1 100 1.0"],  
            special_bonds="lj 1.0 1.0 1.0",    # LAMMPS default: "lj 0 0 0" (excludes 1-2/1-3/1-4)  
            ... )
```

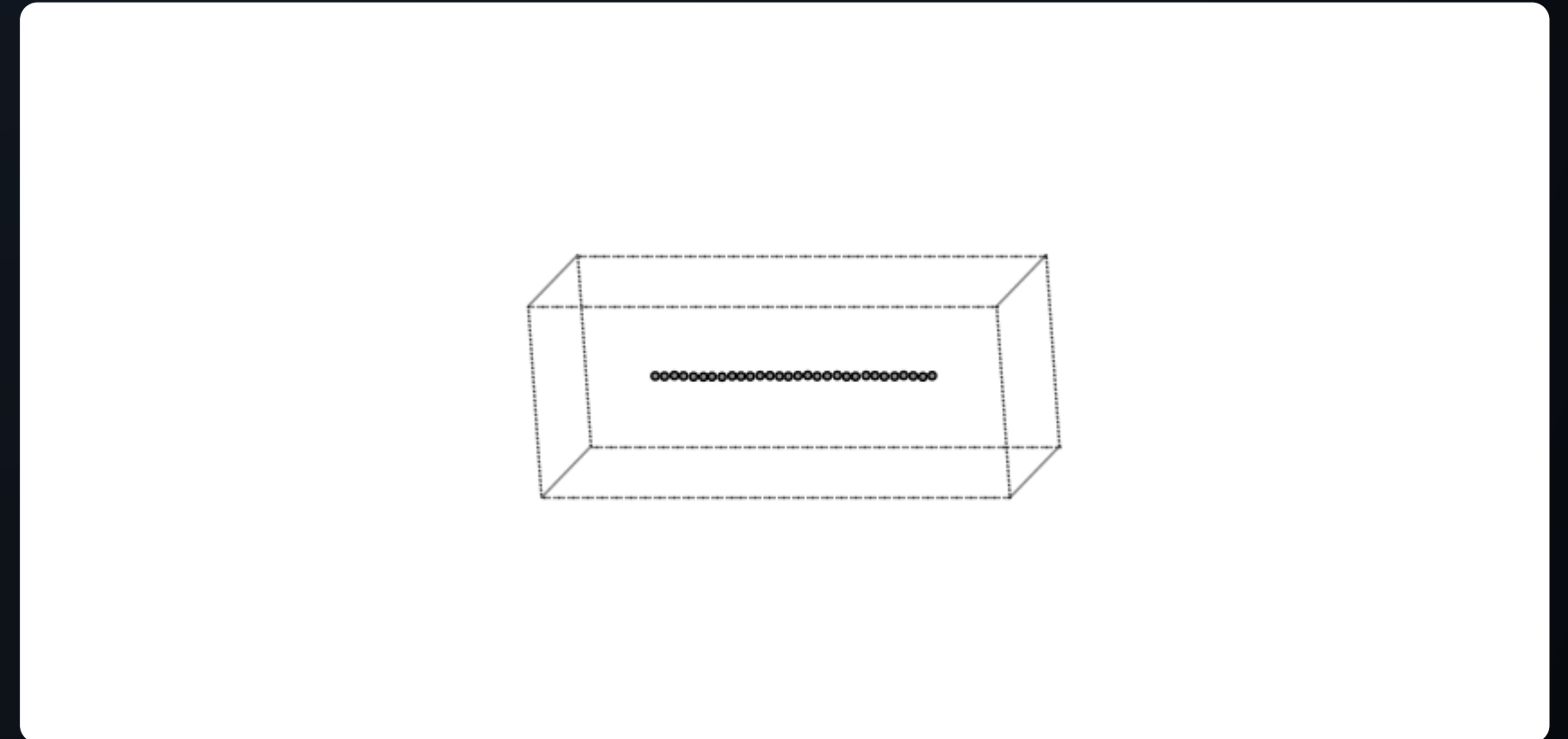
- Without `special_bonds lj 1.0 1.0 1.0`, LAMMPS and mdlite would compute two different systems and disagree for a reason that looks like a bug in either engine -- it is neither, it is an unmatched default.
- Not academic here: this chain's 1-3 neighbours sit around 2σ , well inside the 2.5σ cutoff, so the excluded interaction would be real, not negligible.

Watching the chain contract

The starting conformation (left) and the same chain animated across a real 2000-step run (right) -- it visibly contracts under thermal motion at $T^*=1.0$. How much of that is specifically due to full Lennard-Jones is quantified next, not asserted here.



The chain's starting conformation: near-straight, the small per-bead jitter `bead_spring_chain()` adds barely visible at this scale. Rendered as carbon-like spheres purely to pick a radius and colour -- reduced LJ units carry no real element, the same convention chapter 01's atom views use. – Figure 1 · notebook §0 · chapter 11, cell 9



The same chain animated across the run: the near-straight starting conformation contracting as thermal motion ($T^*=1.0$) samples the space the harmonic bonds and full (attractive+repulsive) Lennard-Jones allow -- the same physical case the energy/force cross-check above verified, seen directly rather than only through a single static configuration. How much of this contraction is specifically because this model uses full LJ rather than WCA is measured, not asserted, next. – Figure 2 · notebook §0 · chapter 11, cell 10

- The picture alone cannot separate 'a flexible chain relaxes from an artificially stretched start' from 'full LJ specifically drives a collapse' -- both would look similar over one run.
- The next slide isolates the difference with a controlled comparison, on the identical chain, in mdlite alone.

Isolating the variable: full LJ against WCA, in mdlite alone

The identical starting chain, the identical step count and thermostat, run twice through mdlite's own `velocity_verlet` -- once with full ($\text{rcut}=2.5$) Lennard-Jones, once with WCA ($\text{rcut}=2^{(1/6)}$, the potential's own minimum, `shift=True`). No LAMMPS-vs-mdlite difference can confound this comparison: both runs are mdlite.

Model	R_g (start)	R_g (end)
Full Lennard-Jones	8.66	2.34
WCA (purely repulsive)	8.66	3.34

- Full LJ ends about 30% more contracted than WCA on the identical chain -- measured, not assumed.
- WCA needed no new mdlite code at all: it is `mdlite.pair.LennardJones` with a different `rcut` and `shift=True`, the same class this lecture uses throughout.

One configuration, mdlite against LAMMPS

The identical chain, evaluated once (no dynamics): mdlite's LennardJones + HarmonicBond against LAMMPS's pair_style lj/cut + bond_style harmonic on the same positions and topology.

Quantity	mdlite	LAMMPS	Agreement
Energy	measured	measured	to 3×10^{-7} (thermo print precision)
Force (max component)	measured	measured	to 2×10^{-11} (round-off)

Water

Rigid SPC/E water: PPPM electrostatics and fix shake, neither in mdlite -- the first chapter validated against a NIST reference instead of a cross-engine check, plus the oxygen-oxygen structure and a real animated box.

- 1 The first chapter with no mdlite comparison
- 2 lammpskill.script.spce_water: already built, used nowhere until now
- 3 Oxygen-oxygen structure: $g_{OO}(r)$
- 4 Watching the water box
- 5 Density against a NIST reference

The first chapter with no mdlite comparison

SPC/E water is a rigid three-site model (bond and angle geometry enforced by fix shake, not integrated as flexible degrees of freedom) held together by long-range Coulomb interactions on top of Lennard-Jones. mdlite implements neither rigid-body constraints nor Ewald/PPPM electrostatics.

- Adding either would be new mdlite physics well beyond this roadmap item's own scope -- the design spec named this explicitly: validation against a published reference, not a new mdlite capability.
- So this chapter's cross-check is against the literature instead: SPC/E's own saturated liquid density at 300 K, from a real NIST reference table -- the same public-domain SRSW-style source this course's own Lennard-Jones fluid benchmark already uses.

lammpskill.script.spce_water: already built, used nowhere until now

A cubic lattice of rigid SPC/E molecules, built and tested before this chapter existed -- the lattice spacing (3.1 Å) was chosen to already sit close to water's real density.

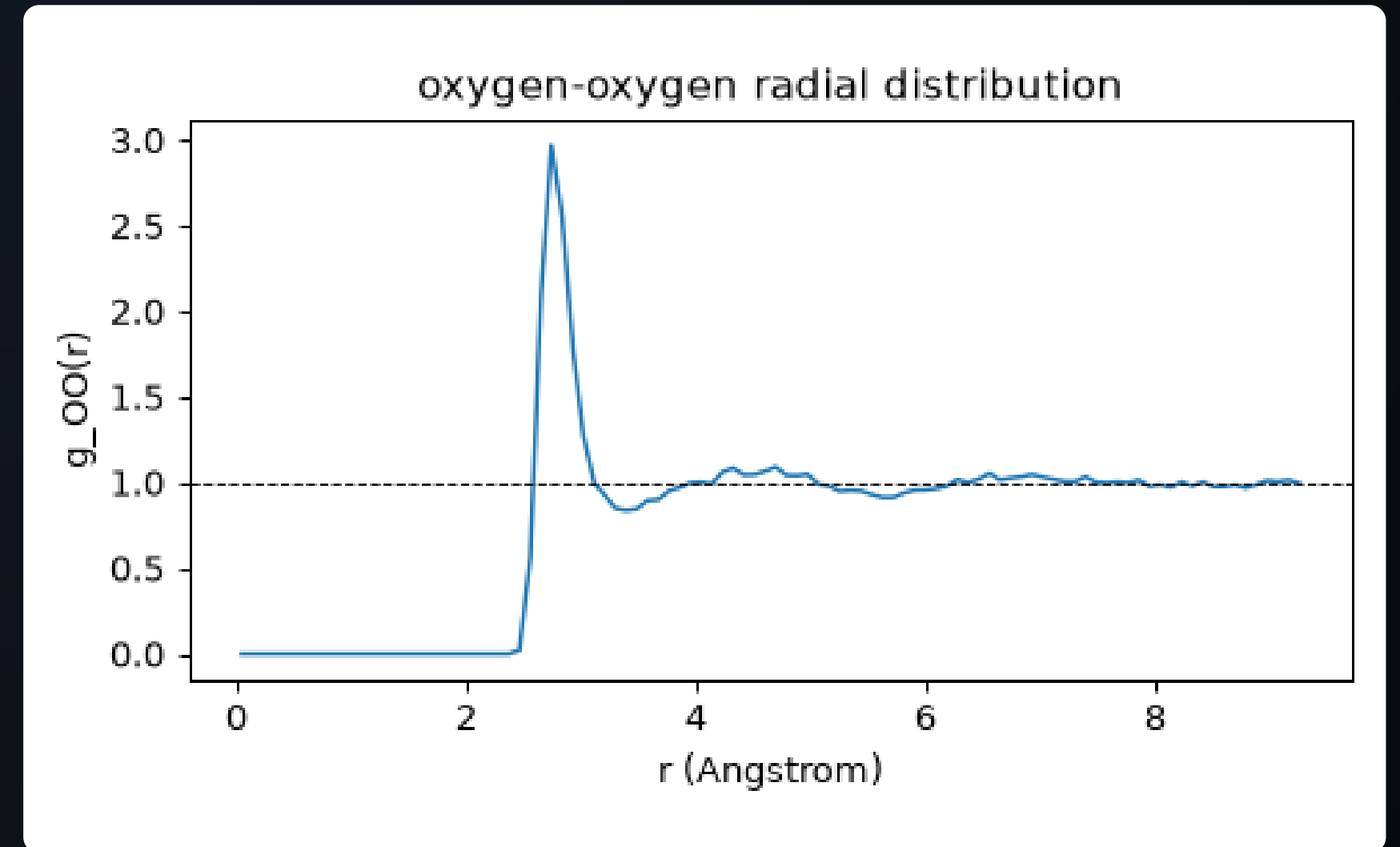
```
spec, df = spce_water(n_side=6)
# fix shk all shake 1.0e-4 20 0 b 1 a 1    -- rigid O-H bonds, H-O-H angle
# kspace_style ppm 1.0e-4                  -- long-range electrostatics
# 216 molecules, 648 atoms, 432 bonds, 216 angles
```

- `check()` carries no error on this preset either -- built and verified the same way as every other case in this course, real molecular complexity included.

Oxygen-oxygen structure: $g_{OO}(r)$

The box's own radial distribution function, from a real trajectory (2.5 ps of relaxation, not a long equilibration) -- a sharp first peak near the hydrogen-bonding distance and a modest second-shell feature near 4.5 Å, decaying to the ideal-gas value of 1.

- First peak measured here: $r=2.74$ Å, $g=2.97$ -- for scale, not a pass/fail check, published SPC/E values place it around 2.7-2.8 Å (Mark & Nilsson 2001; Camisasca et al. 2019, comparing SPC/E directly to experimental X-ray diffraction).
- Computed by `lammppskill.post.rdf_trajectory` -- the same tool chapter 06 uses, on a real molecular system instead of a monatomic LJ fluid.

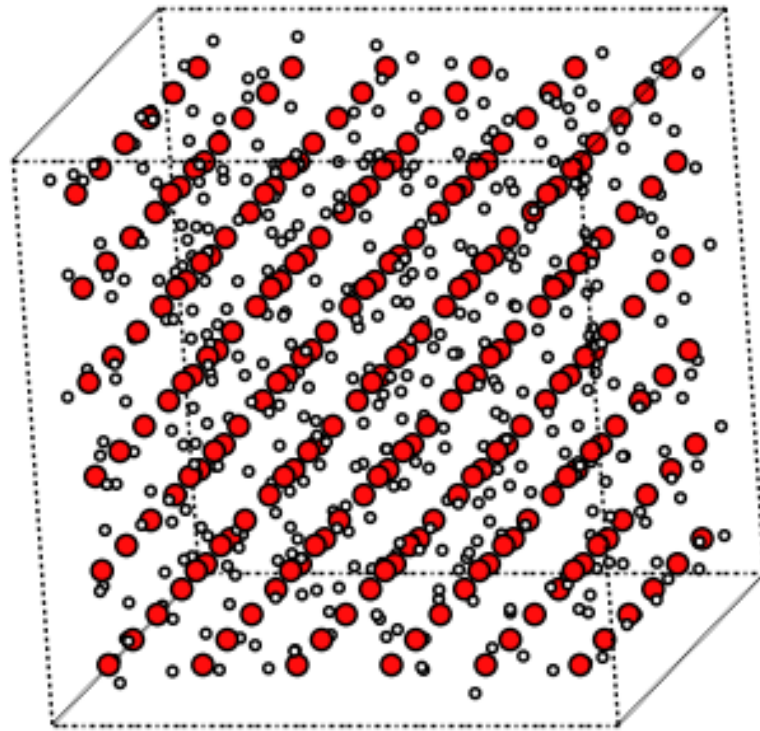


The oxygen-oxygen radial distribution function from this box's own trajectory, after 2.5 ps of relaxation off the lattice start: a first peak near the O-O hydrogen-bonding distance, decaying to the ideal-gas value of 1 at large r -- computed by `lammppskill.post.rdf_trajectory`, the same tool chapter 06 uses.

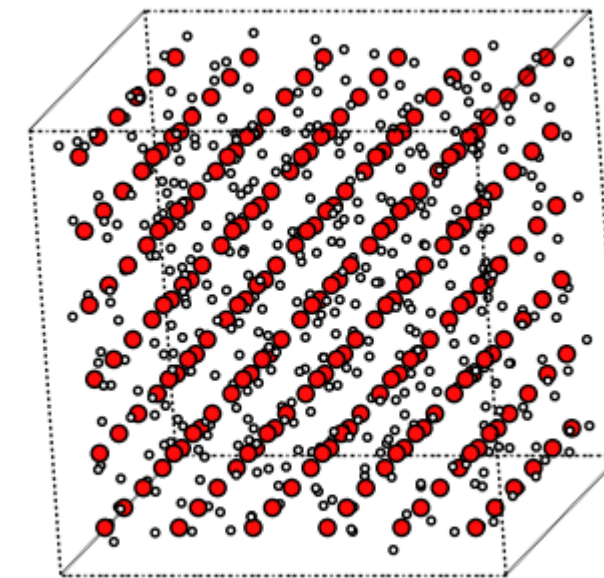
– Figure 1 · notebook §0 · chapter 12, cell 10

Watching the water box

The starting configuration (left) and the same box animated across a real 5000-step run (right) -- real atomic masses (15.9994, 1.008), so the mass-to-symbol guess renders oxygen and hydrogen directly, unlike every earlier reduced-unit chapter.



The SPC/E water box near the start of the run -- real atomic masses (15.9994, 1.008), so lammps.kitviz's own mass-to-symbol guess correctly renders oxygen and hydrogen, unlike every earlier reduced-unit chapter, which had to pick a conventional stand-in element by hand. — Figure 2 · notebook §0 · chapter 12, cell 11



The same water box animated across the run: rigid molecules (fix shake) jiggling and reorienting under thermal motion, the same physical case the density and $g_{OO}(r)$ checks above both drew from.

— Figure 3 · notebook §0 · chapter 12, cell 12

- The visible lattice order at the start disorders under thermal motion as fix shake holds each molecule rigid while it reorients -- the same physical case (fix nvt, not the density check's fix npt) the $g_{OO}(r)$ plot was also computed from.
- This run is separate from the density benchmark above: a second real run, for structure and the atom views, not part of either pass/fail check.

Density against a NIST reference

LAMMPS's own fix npt at 300 K / 1 atm (chapter 04's fix npt pattern, its first use on a real molecular system), block-averaged over a post-equilibration tail, against NIST's SAT-TMMC saturated liquid density at 300 K.

Quantity	LAMMPS (fix npt)	NIST (SAT-TMMC)	Agreement
Density (kg/m3)	994.0 ± 3.8	998.1 ± 2.9	within 20.2 (S4: 3σ combined)